CLASS 9 — MATHS · FORMULA SHEET

Chapter 2: **Linear Polynomials** | Quick revision, one page

Linear polynomial

- A linear polynomial in x has the form $p(x) = ax + b$ with $a \neq 0$; its degree is 1.
- Value of $p(x)$ at $x = k$: replace x by k, i.e. $p(k) = ak + b$.
- Zero of $p(x)$: the value k with $p(k) = 0$; for $ax + b$ it is $k = -\dfrac{b}{a}$. A linear polynomial has exactly one zero.

Graph of $y = ax + b$

- The graph is a straight line.
- Slope $= a$ (the coefficient of x): the change in y for a change of 1 in x.
- y-intercept $= b$: the line meets the y-axis at $(0, b)$.
- x-intercept $= -\dfrac{b}{a}$: the line meets the x-axis at $\left(-\dfrac{b}{a}, 0\right)$ (the zero of the polynomial).
- $a > 0$: the line rises to the right; $a < 0$: it falls.

Nth term and real-life rates

- If a sequence increases by a constant d each time and starts at t_1, then $t_n = dn + (t_1 - d)$, a linear polynomial in n.
- Total $=$ (rate) $\times$ (quantity) $+$ fixed amount; the rate is the slope and the fixed amount is the y-intercept.
- To find the input for a given output, solve $ax + b = \text{output}$.

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